Pearson Edexcel International A Level · Pure Mathematics 3

# Chapter 2 — Functions and graphs
# Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第2章 · 函数与图像
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Figure 1 gives the complete graph of $y = f(x)$, formed from straight line segments joining $(-4, 1)$, $(-1, 4)$, $(2, -2)$ and $(5, 1)$. All four endpoints are included.

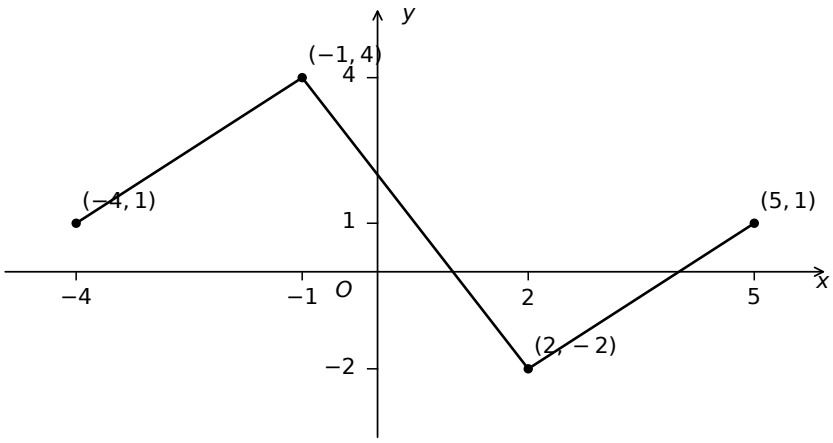



Figure 1

(a) Find $ff(2)$ and all solutions of $f(x) = 1$. (3)

(b) Let $g(x) = 2x - 1$ for real $x$. Find the domain and range of $fg$. (3)

(c) Sketch $y = |f(x)|$, marking every zero and every corner point. (3)

(Total for Question 1 is 9 marks)

The function $f(x) = 4 - (x - 1)^2$ has domain $[-1, 4]$. Define $g(x) = |f(x)|$ on the same domain, and $h(x) = f(|x|)$ wherever this is defined. Figure 2 shows the original graph.

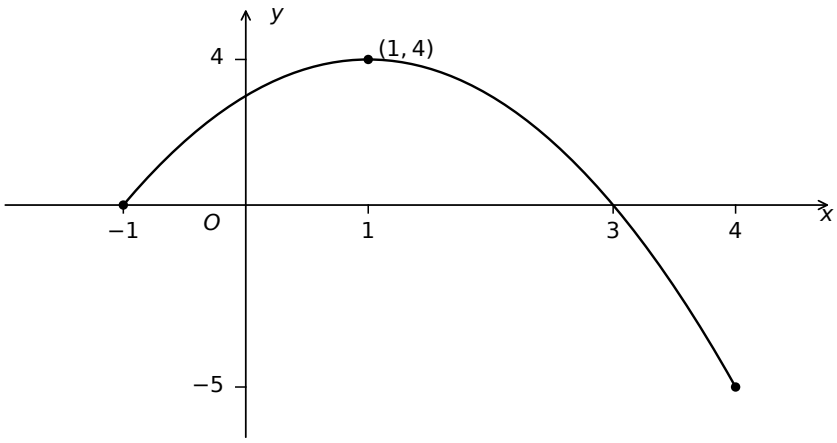



Figure 2

(a) Find the domain and range of $h$ and sketch its graph, labelling its zeros and maximum points. (4)

(b) Find the range of $g$. (2)

(c) Find all $k$ for which $y = g(x)$ and $y = k$ have exactly two distinct intersections. (3)

(Total for Question 2 is 9 marks)

An instrument maps a non-negative input $x$ to the reading $f(x) = (ax + b)/(x + 2)$. Its calibration graph has horizontal asymptote $y = 3$ and passes through $(2, 2)$, as shown in Figure 3.

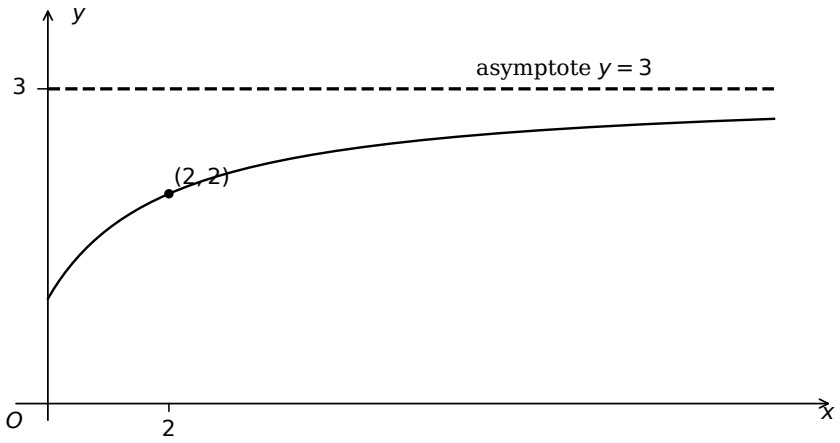



Figure 3

(a) Find $a, b$ and write $f(x)$ in the form $3 - c/(x + 2)$. (3)

(b) Find the range of $f$ and the inverse function, with its domain. (4)

(c) Find the input for which the reading is $0.2$ below the limiting reading. Explain why the limiting reading itself is never attained. (3)

(Total for Question 3 is 10 marks)

**4.**



Let $f(x) = \sqrt{x - 1}$ for $x \geq 1$, and $g(x) = x^2 + 1$ for $x \geq 0$.

(a) Show that $f$ and $g$ are inverse functions, stating the domain of each composition. (4)

(b) The domain of $g$ is now extended to all real numbers, while $f$ is unchanged.

(i) Explain why $g$ does not have an inverse function on this extended domain. (2)

(ii) Solve $fg(x) = 3$ for real $x$. (3)

(Total for Question 4 is 9 marks)

Figure 5 shows the complete graph of $f(x) = a|x - b| + c$ on $[-2, 6]$, where $a > 0$. Its vertex is $(2, -3)$ and its zeros are $-1, 5$. The two endpoints are included.

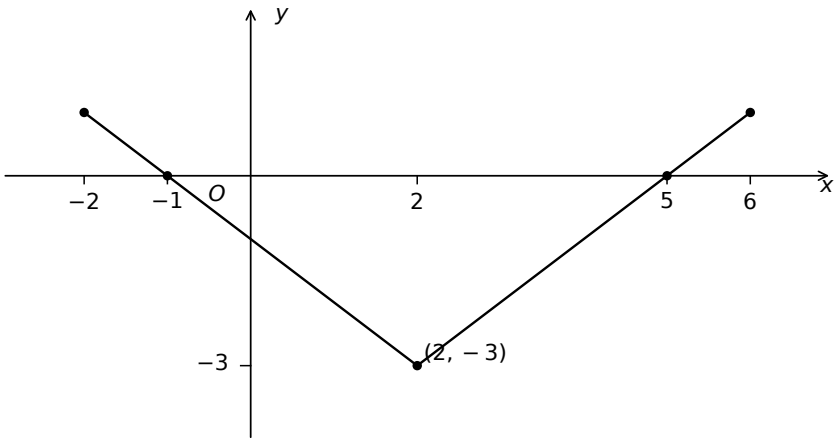



Figure 5

(a) Find $a, b, c$. (3)

(b) Find all $k$ such that $f(x) + k = 0$ has two distinct solutions in the stated domain. (3)

(c) Solve $f(|x|) \geq 1$, respecting the domain of the composition. (3)

(Total for Question 5 is 9 marks)

A depth sensor in a tank of height 4 m gives reading $s = f(x) = (4 - x)^2$, where $x$ m is the water depth and $0 \leq x \leq 4$. Figure 6 shows the calibration graph.

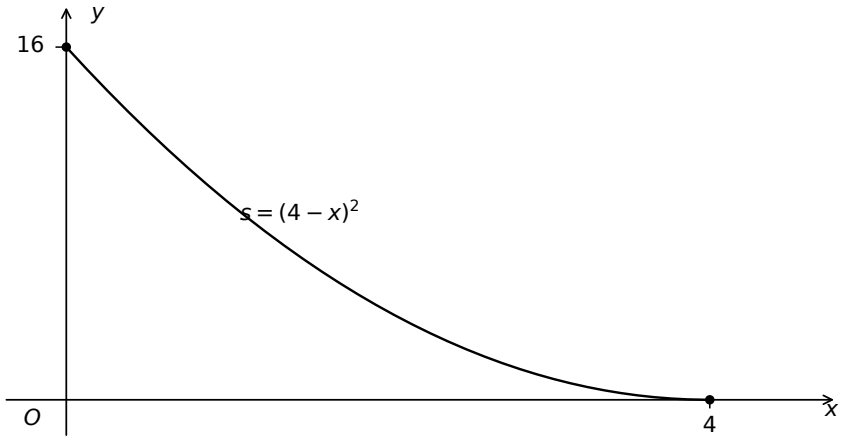



Figure 6

(a) State the range of the reading and find the depth corresponding to reading $9$. (4)

(b) Find the inverse calibration function, with its domain. Explain the choice of square-root sign. (3)

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \leq x \leq 8$.

(i) Find $h^{-1}(s)$, stating its domain. (3)

(ii) Find the depth corresponding to reading $s = 9$. (2)

(Total for Question 6 is 12 marks)

**7.**



Let $f(x) = 1/(x-2)$ for $x > 2$, and $g(x) = \sqrt{x+1}$ for $x \geq -1$.

(a) Find $fg(x)$ and its domain. (3)

(b) Find $gf(x)$ and its domain and range. (4)

(c) Solve $fg(x) = 1$ and $gf(x) = \sqrt{3}$, giving each answer separately. (4)

(Total for Question 7 is 11 marks)

Figure 8 shows the complete graph of $f$ on $[-2, 6]$. Its maximum is $f(0) = 5$, its minimum is $f(3) = -1$, and all points between these output values occur. Define $h(x) = -\frac{1}{2}f(2x) + 3$.

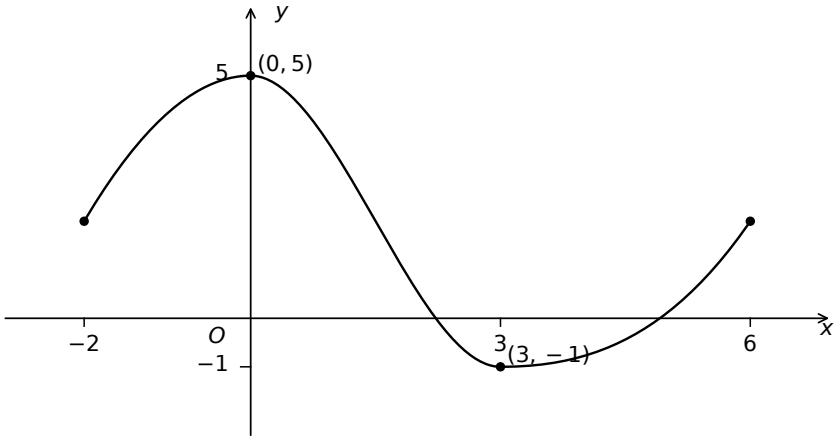



Figure 8

(a) Describe a sequence of transformations from the graph of $f$ to that of $h$. (3)

(b) Find the domain, range and both extreme points of $h$. (3)

(c) Find the domain and range of $f(|x|)$. Explain why the range is not lost by discarding the negative-input part of $f$. (3)

(Total for Question 8 is 9 marks)

Let $f(x) = (x+1)/(x-1)$ for real $x \neq 1$.

(a) Find $f^{-1}$ with its domain, and hence find $ff(x)$. (4)

(b) Using this function, give a counterexample to the statement that a function graph and its inverse graph can intersect only on $y = x$. (3)

(c) Find the points where the common graph does meet $y = x$. (2)

(Total for Question 9 is 9 marks)

The function is defined by $f(x) = x + 2$ for $-2 \leq x \leq 1$, and $f(x) = 5 - x$ for $1 < x \leq 4$. Figure 10 shows the open and closed endpoints.

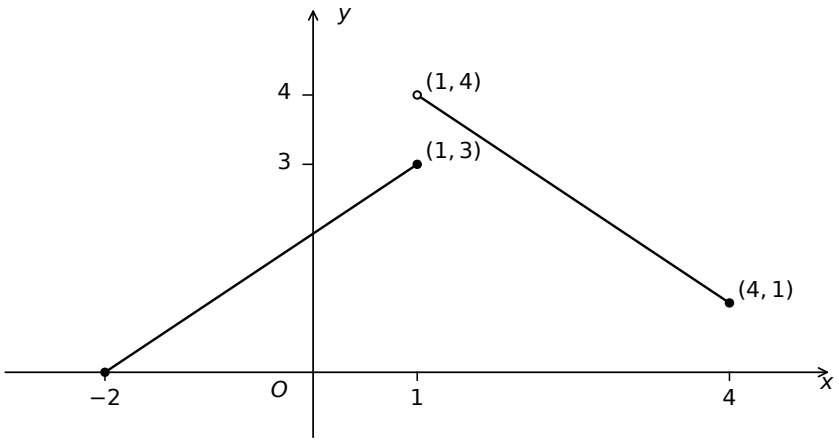



Figure 10

(a) Find the range of $f$ and $ff(1)$. (3)

(b) Does $f$ attain a greatest value? Explain using the endpoint at $x = 1$. (2)

(c) Restrict the domain to $[-2, 3/2]$. Find the inverse function in piecewise form, with its complete domain. (4)

(Total for Question 10 is 9 marks)

Let $f(x) = |x - 3|$ and $g(x) = |x| - 3$, both on the real numbers.

(a) Find $fg(x)$ and $gf(x)$, simplifying the outer absolute value where possible. (3)

(b) Solve $fg(x) = gf(x)$ for real $x$. (4)

(c) Use the input $x = 0$ to verify that the two composite functions are not identical. (2)

(Total for Question 11 is 9 marks)

Let $g(x) = 3 - \sqrt{x + 2}$ for $-2 \leq x \leq 7$. Figure 12 shows its graph.

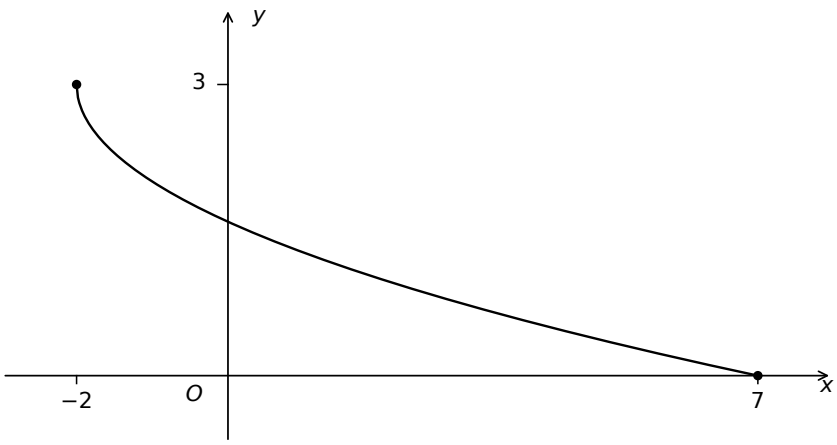



Figure 12

(a) Find its range and inverse function, with the domain of the inverse. (4)

(b) Solve $gg(x) = 1$ without expanding a nested square root. (3)

(c) Explain why $gg$ is defined for every $x$ in the original domain. (2)

(Total for Question 12 is 9 marks)

Let $f(x) = \sqrt{9 - x^2}$ on $[-3, 3]$, and $g(x) = x^2$ for real $x$.

(a) Find $fg(x)$ and its full domain and range. (4)

(b) Find $gf(x)$ and its full domain and range. (3)

(c) Solve $fg(x) = \sqrt{5}$ and $gf(x) = 5$, giving both solution sets. (2)

(Total for Question 13 is 9 marks)

The graph in Figure 14 is $y = f(x) = |x^2 - 4|$ on $[-3, 3]$. Define $h_k(x) = f(x) + k$ on this domain.

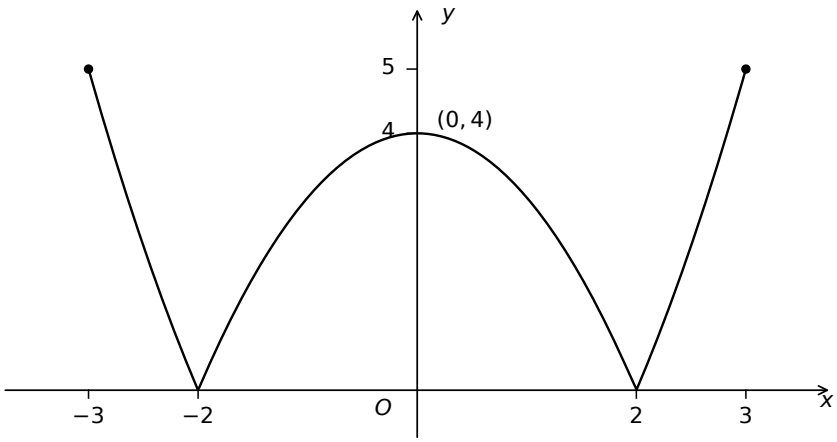



Figure 14

(a) Find the range of $f$ and identify every point where the minimum occurs. (3)

(b) Find all $k$ for which $h_k$ has exactly four distinct zeros. (3)

(c) Find the value of $k$ for which $h_k$ has exactly three distinct zeros, and give these zeros exactly. (3)

(Total for Question 14 is 9 marks)

Let $q(x) = (x^2 - 9)/(x - 3)$ for real $x \neq 3$.

(a) Sketch the graph of $q$, showing the missing point. State its range. (3)

(b) Find $qq(x)$ with its complete domain and range. (4)

(c) Find $q^{-1}$ with its domain. (2)

(Total for Question 15 is 9 marks)

Figure 16 shows $f(x) = x^3 - 3x$ on $[-2, 2]$. Its turning points are $(-1, 2)$ and $(1, -2)$, and its endpoints are $(-2, -2), (2, 2)$.

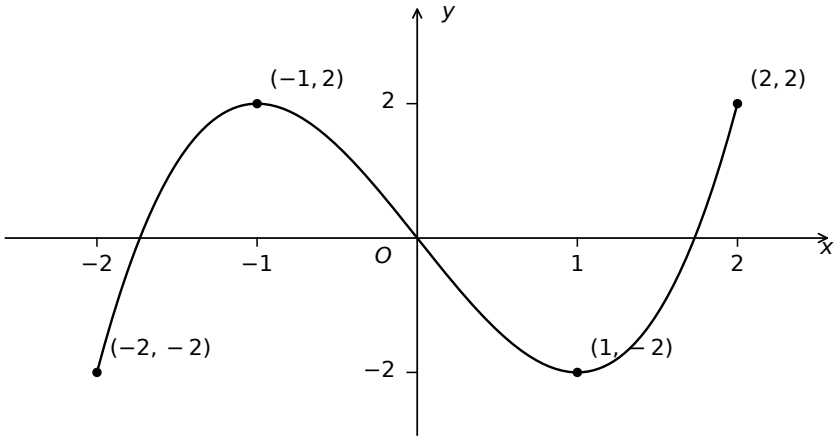



Figure 16

(a) Find all values of $k$ for which $f(x) = k$ has exactly two distinct solutions. (3)

(b) Sketch $y = |f(x)|$, marking every zero and every point at height $2$. (3)

(c) For which $k$ does $|f(x)| = k$ have six distinct solutions? (3)

(Total for Question 16 is 9 marks)

A calibration is $f(x) = 18/(x+3)$ for $0 \le x \le 6$. A correction procedure adds 1 to a reading and then uses the inverse calibration. Its output is $H(x) = f^{-1}(f(x)+1)$ whenever this is defined.

(a) Find the range of $f$ and $f^{-1}$, including its domain. (4)

(b) For the correction procedure $H(x) = f^{-1}(f(x)+1)$,

(i) Find the complete domain of $H$. (3)

(ii) Simplify $H(x)$ and find the input for which $H(x) = 1$. (4)

(Total for Question 17 is 11 marks)

Figure 18 gives a smooth graph $y = f(x)$ on $[-3, 4]$. Its only zeros are $-2, 1, 3$; its minimum is $(0, -2)$ and maximum is $(2, 4)$. The endpoint values are $f(-3) = 2$ and $f(4) = -1$. Define $h(x) = -2f(x/3) - 1$.

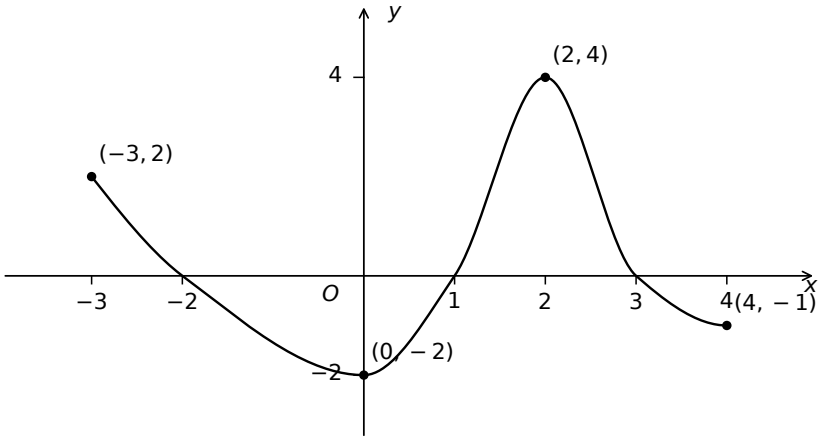



Figure 18

(a) Find the domain and range of $h$. (4)

(b) Find the maximum and minimum points of $h$. (3)

(c) Find all exact solutions of $h(x) = -1$ using the given graph information. (3)

(Total for Question 18 is 10 marks)

Two depots lie at positions $1$ and $4$ km on a straight road. Travelling one kilometre to the first depot has cost $1$ unit; the corresponding cost to the second depot is $2$ units. A delivery point at $x$ has total cost $F(x) = |x - 1| + 2|x - 4|$.

(a) Write $F$ in piecewise form and sketch its graph. (3)

(b) Solve $F(x) \le 6$. (2)

(c) Restrict the delivery point to $x \ge 4$. Find the inverse cost function, with its domain. (3)

(Total for Question 19 is 8 marks)

Let $f(x) = \sqrt{16 - (x-1)^2}$ for $1 \leq x \leq 5$. Figure 20 shows its graph.

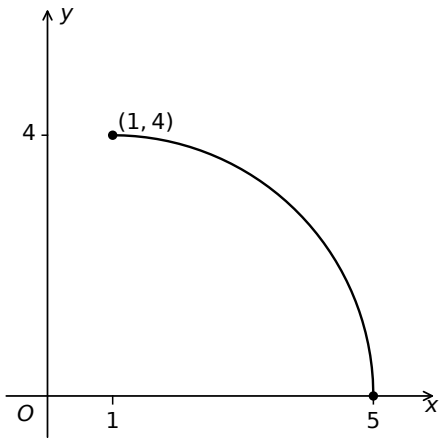



Figure 20

(a) Find $f^{-1}$, stating its domain and range. (4)

(b) Find all intersections of the two graphs exactly. Justify why none lies off $y = x$ for this particular pair. (4)

(Total for Question 20 is 8 marks)

The function $f(x) = |2^x - 4|$ is defined for all real $x$.

(a) Sketch the graph, labelling both intercepts and its horizontal asymptote, and state its range. (4)

(b) Find all values of $k$ for which $f(x) = k$ has two distinct real solutions. (3)

(c) Solve $f(x) = 3$ exactly. (2)

(Total for Question 21 is 9 marks)

Let $f(x) = |1/x - 1|$ for real $x \neq 0$.

(a) Sketch the graph, showing all asymptotes and its zero, and state the range. (4)

(b) Find all $k$ such that $f(x) = k$ has exactly two distinct solutions. (3)

(c) Solve $f(x) > 2$. (3)

(Total for Question 22 is 10 marks)

Let $f(x) = x^2 - 4x + 7$ for $x \geq 2$, and define $g(x) = f(x) - x$ on the same domain.

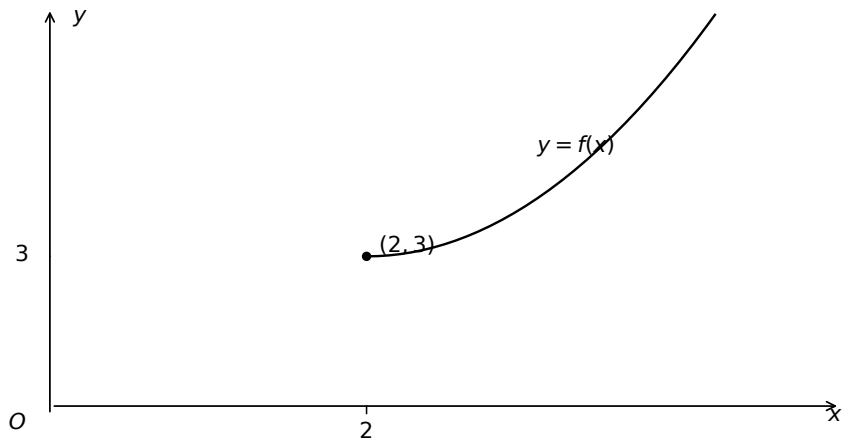



Figure 23

(a) Find $f^{-1}$ with its domain. (3)

(b) Show that $g$ is not one-to-one on its full domain, and find its range. (3)

(c) Find both inverses obtained by restricting $g$ to its maximal one-to-one intervals within $[2, \infty)$. Include the domain of each inverse. (4)

(Total for Question 23 is 10 marks)

The complete graph of $f$ is the two straight segments joining $(-2, -1)$ to $(0, 2)$ to $(2, -1)$, as shown in Figure 24.

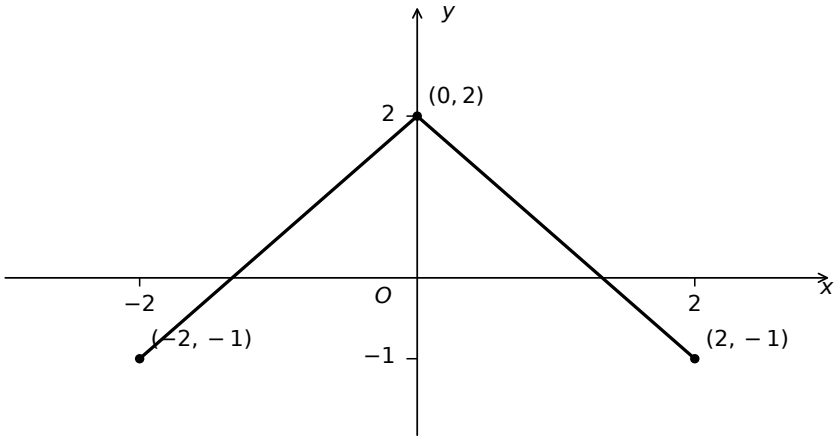



Figure 24

(a) Write an expression for $f$ and state its range. (3)

(b) Restrict $f$ to $[0, 2]$ and find the inverse of this restriction, including its domain. (3)

(c) Using the original full-domain $f$, find every solution of $ff(x) = 0$. (4)

(Total for Question 24 is 10 marks)

A two-step calibration applies $f(x) = (x+2)/(x+4)$ for $x \geq 0$, followed by $g(s) = 2s - 1$ for $1/2 \leq s < 1$. The complete calibration is $h = gf$.

(a) Find $h(x)$ and its range. (3)

(b) The complete calibration $h$ is used to recover the original non-negative input from an output reading $r$.

(i) Find $h^{-1}(r)$ and state its domain. (3)

(ii) Find the input that gives reading $r = 1/4$. (2)

(c) A technician attempts to undo $f$ first and then $g$. Explain the error using the reading $1/4$. (3)

(Total for Question 25 is 11 marks)

Let $f(x) = \dfrac{x-2}{x+1}$ for $x > 2$ and $g(x) = \dfrac{1}{x}$ for $x > 0$.

(a) Find the range of $f$ and its inverse, including the domain of the inverse. (4)

(b) For the composite $gf$, find

(i) its expression and full domain, (3)

(ii) its range. (2)

(Total for Question 26 is 9 marks)

$f(x) = |x - 2| + 1$ is defined for $x \geq 2$. Define $h(x) = f(3 - x)$ wherever this is valid.

(a) Find $f^{-1}$ and state its domain. (3)

(b) For $h$, find

(i) its domain and simplified expression, (3)

(ii) $h^{-1}$ with its domain and range. (3)

(Total for Question 27 is 9 marks)

The function $f$ is defined by $f(x) = x + 1$ for $0 \leq x < 2$, and $f(2) = 5$.

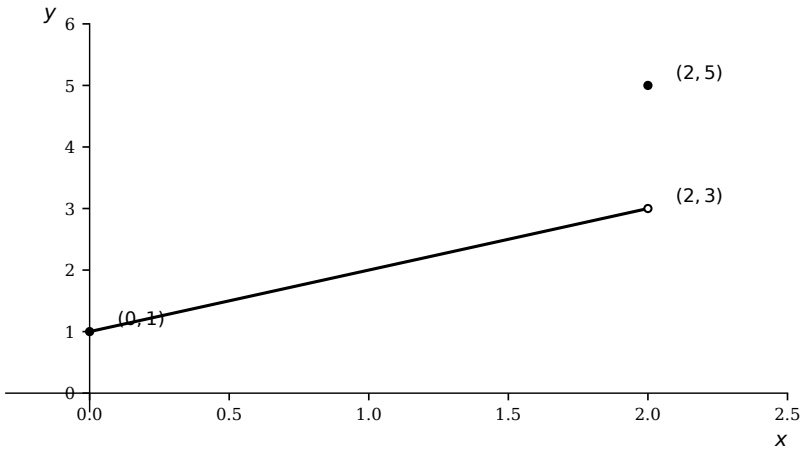



Figure 28

(a) State its range and determine whether it is one-to-one. (3)

(b) For the inverse function, find

(i) its expression on its complete domain, (3)

(ii) all $x$ for which $ff(x)$ is defined. (3)

(Total for Question 28 is 9 marks)

The function $f$ is defined by $f(x) = |2x - 6| + 1$ for $0 \leq x \leq 5$.

(a) Sketch the graph of $f$, marking the vertex and both endpoints. State its range. (3)

(b) The domain is now restricted to $k \leq x \leq 5$, where $0 \leq k < 5$, so that $f$ has an inverse.

(i) Find the smallest possible value of $k$. (2)

(ii) For this restriction, find $f^{-1}(x)$ and its domain. (3)

(Total for Question 29 is 8 marks)

The functions $f$ and $g$ are defined by $f(x) = \sqrt{x+2}$ for $x \geq -2$, and $g(x) = \dfrac{1}{x-1}$ for $x \neq 1$.

(a) Find $gf(x)$ and state its full real domain. (3)

(b) For the composite function $fg$,

(i) find the set of inputs for which it is defined. (3)

(ii) solve $fg(x) = \sqrt{3}$. (2)

(Total for Question 30 is 8 marks)

# Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

## Question 1

P3-C02-R01 · 9 marks

### (a) · 3 marks

$ff(2)=3$; $x=-4, 1/2, 5$.

f(2)=−2，在第一线段f(−2)=3。水平线y=1与首点、中央线段x=1/2以及末点相交。

| | |
|---|---|
| **M1** | 按顺序从图读取复合值 |
| **A1** | ff(2) |
| **A1** | 三个有效横坐标 |

### (b) · 3 marks

Domain $[-3/2, 3]$; range $[-2, 4]$.

要求2x−1∈[−4,5]，解得新区间。g在此区间恰好覆盖f整个定义域，所以复合值域与f相同。

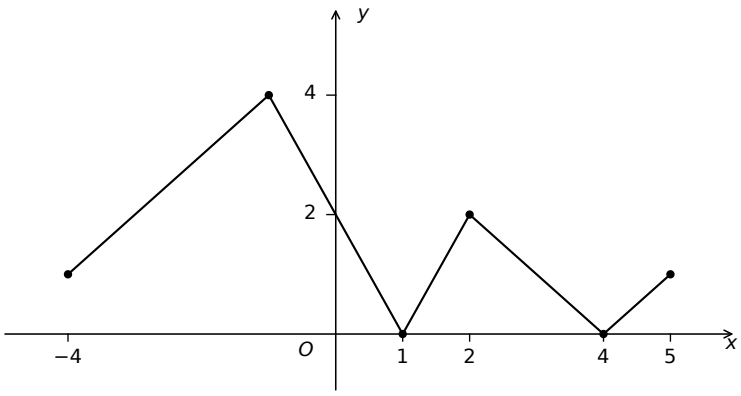
| | |
|---|---|
| **M1** | 内层输出必须落在外层定义域 |
| **A1** | 定义域 |
| **B1** | 完整值域及覆盖理由 |

### (c) · 3 marks

Corner points $(-4,1), (-1,4), (1,0), (2,2), (4,0), (5,1)$; zeros $1, 4$.

保持横轴以上原图；把中央后段和最后前段的负值部分向上反射。中央线段零点1、末线段零点4，原低点(2,−2)变(2,2)。

| | |
|---|---|
| **M1** | 只反射原图负值部分 |
| **A1** | 两个零点 |
| **A1** | 其余全部角点和正确连线 |




Suggested sketch

## Question 2

P3-C02-R02 · 9 marks

### (a) · 4 marks

Domain $[-4,4]$, range $[-5,4]$; zeros $x=\pm 3$; maxima $(\pm 1,4)$; $h(0)=3$.

|x|只取原定义域中的$[0,4]$，再将这一半关于纵轴反射。不能把原图x<0的部分也保留。右端值−5仍是最小值。

| | |
|---|---|
| **M1** | 选取原图非负横坐标部分并反射 |
| **A1** | 定义域和值域 |
| **A1** | 两零点 |
| **A1** | 最大点及拼接点 |

### (b) · 2 marks

$[0,5]$.

原函数最小−5、最大4且过轴；负值绝对值后能到5，故新最大值不是4。

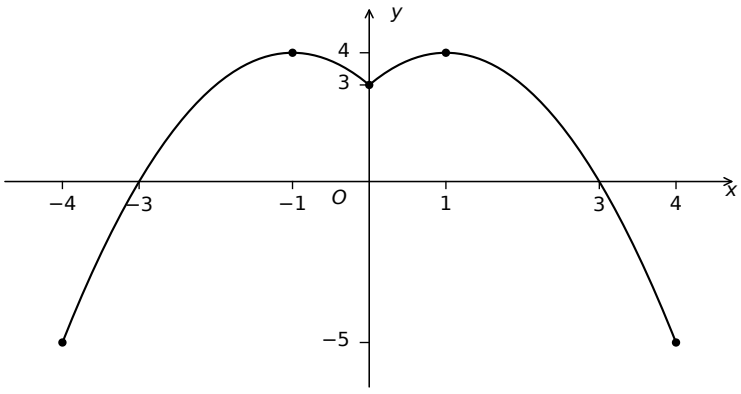
| | |
|---|---|
| **M1** | 比较原正最大值与负最小值绝对值 |
| **A1** | 完整值域 |

### (c) · 3 marks

$k=0$ or $k=4$.

k=0时原零点−1、3两点；0

| | |
|---|---|
| **M1** | 按图分高度区间并处理极值 |
| **A1** | k=0 |
| **A1** | k=4且无额外值 |




Suggested sketch

## Question 3

P3-C02-R03 · 10 marks

### (a) · 3 marks

$a = 3, b = 2$; $f(x) = 3 - 4/(x + 2)$.

大x时分子分母首项比给a=3；点(2,2)给6+b=8。再用除法整理成3−4/(x+2)。

| | |
|---|---|
| **B1** | 从渐近线确定a |
| **M1** | 代点求b |
| **A1** | 参数与混合式 |

### (b) · 4 marks

Range $[1, 3)$; $f^{-1}(x) = (2x - 2)/(3 - x)$, domain $1 \le x < 3$.

对x⩾0，4/(x+2)在(0,2]，故f在[1,3)。由y(x+2)=3x+2解出原输入，交换变量并用原值域作为逆函数定义域。

| | |
|---|---|
| **M1** | 用混合式求值域端点 |
| **A1** | 完整半开值域 |
| **M1** | 反解方程 |
| **A1** | 逆函数及其定义域 |

### (c) · 3 marks

$x = 18$; $f(x) < 3$ for every finite $x \ge 0$.

与渐近线的差为4/(x+2)，令为0.2给x+2=20。对有限非负x这项严格为正，故无法正好输出3。

| | |
|---|---|
| **M1** | 用与极限的差建立方程 |
| **A1** | 输入18 |
| **B1** | 不能达到渐近线的代数理由 |

## Question 4

P3-C02-R04 · 9 marks

(a) · 4 marks

$fg(x) = x$ on $[0, \infty)$; $gf(x) = x$ on $[1, \infty)$.

fg=√(x²)，因g的输入x⩾0可写x；gf=(√(x−1))²+1=x，需x⩾1。两个恒等式均必须带各自输入范围。

| | |
|---|---|
| **M1** | 正确计算fg并处理根号平方 |
| **A1** | fg及范围 |
| **M1** | 正确计算gf |
| **A1** | gf及范围 |

(b) The domain of $g$ is now extended to all real numbers, while $f$ is unchanged.

(i) · 2 marks

$g(u) = g(-u)$ for every $u > 0$, so it is many-to-one.

正负同值输入输出相同，例如g(2)=g(−2)=5；逆映射将给同一输入两个输出，不能是函数。

| | |
|---|---|
| **M1** | 有效重复输出例子或一般关系 |
| **A1** | 非一一映射与无逆函数的解释 |

(ii) · 3 marks

$x = \pm 3$ because $fg(x) = |x|$.

√(x²)在实数域必须是|x|。扩展g允许负输入，因此−3不再受第一问域限制。

| | |
|---|---|
| **M1** | 识别全实数域下根号平方为绝对值 |
| **A1** | 两个解 |
| **B1** | 负解与域扩展的联系 |

## Question 5

P3-C02-R05 · 9 marks

### (a) · 3 marks

$a = 1, b = 2, c = -3.$

顶点直接给b=2、c=−3；代零点−1得3a−3=0。

| | |
|---|---|
| **B1** | 顶点确定b,c |
| **M1** | 用一个零点定a |
| **A1** | 参数 |

### (b) · 3 marks

$-1 \le k < 3.$

解|x−2|=3−k。两不同根要求3−k>0；有限域距顶点左右最多都是4，故3−k⩽4。端点可取，顶点不能重复计为两解。

| | |
|---|---|
| **M1** | 用距离表达两根 |
| **M1** | 同时检查不同根和域端点 |
| **A1** | 正确半开区间 |

### (c) · 3 marks

$x = -6$ or $x = 6.$

内层|x|必须落在[−2,6]，故|x|⩽6。再解||x|−2|⩾4：只有|x|⩾6可行，与前域结合得|x|=6。

| | |
|---|---|
| **M1** | 复合定义域 |
| **M1** | 解嵌套绝对值不等式 |
| **A1** | 仅两个端点 |

## Question 6

P3-C02-R06 · 12 marks

### (a) · 4 marks

$0 \le s \le 16$; depth 1 m.

4−x从4降至0，其平方取遍[0,16]；(4−x)²=9候选x=1、7，物理域只允许1。

| | |
|---|---|
| **M1** | 由物理域求输出范围 |
| **A1** | 完整值域 |
| **M1** | 解平方并检查物理域 |
| **A1** | 深度 |

### (b) · 3 marks

$f^{-1}(s) = 4 - \sqrt{s}$, domain $[0, 16]$.

原模型中4−x≥0，因此4−x=√s。若取x=4+√s，除s=0外会超过 tank高度4；反函数域须是可实现读数范围。

| | |
|---|---|
| **M1** | 从原物理域选择根号正负 |
| **A1** | 正确逆函数 |
| **B1** | 逆函数定义域与解释 |

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \le x \le 8$.

#### (i) · 3 marks

$h^{-1}(s) = 8 - 2\sqrt{s}$, $0 \le s \le 16$.

s=(4−x/2)²，由输入范围4−x/2≥0，反解x=8−2√s。

| | |
|---|---|
| **M1** | 正确反解及分支 |
| **A1** | 反函数 |
| **B1** | 定义域 |

#### (ii) · 2 marks

2 m.

代入逆校准8−2×3。

| | |
|---|---|
| **M1** | 代入同一传感器反函数 |
| **A1** | 深度 |

## Question 7

P3-C02-R07 · 11 marks

### (a) · 3 marks

$fg(x) = 1/(\sqrt{x+1} - 2)$; domain $x > 3$.

g输入先需x⩾−1，还要g(x)>2，故√(x+1)>2，最后得x>3。只排除分母为0还不够，因为f本身域为x>2。

| | |
|---|---|
| **M1** | 正确复合顺序 |
| **M1** | 检查外函数的完整输入域 |
| **A1** | 表达式和严格域 |

### (b) · 4 marks

$gf(x) = \sqrt{1 + 1/(x-2)}$; domain $(2, \infty)$, range $(1, \infty)$.

f输入必须>2，其输出遍历正实数，全部可送入g；g对正输入的输出严格>1且无上界。

| | |
|---|---|
| **M1** | 正确复合 |
| **A1** | 域 |
| **M1** | 跟踪内层值域到外层 |
| **A1** | 值域 |

### (c) · 4 marks

$x = 8$ and $x = 5/2$, respectively.

第一式给√(x+1)=3，x=8；第二式平方得1+1/(x−2)=3，x=5/2。分别检查其复合定义域。

| | |
|---|---|
| **M1** | 解第一复合并检查域 |
| **A1** | 第一解 |
| **M1** | 解第二复合并检查域 |
| **A1** | 第二解 |

## Question 8

P3-C02-R08 · 9 marks

### (a) · 3 marks

Horizontal scale factor $1/2$; reflection in the $x$-axis; vertical scale factor $1/2$; translation by $(0, 3)$.

内层2x把横坐标减半；外层−½把纵坐标反号并减半，最后纵向加3。

| | |
|---|---|
| **B1** | 横向尺度 |
| **B1** | 反射及纵向尺度 |
| **B1** | 最后的平移 |

### (b) · 3 marks

Domain $[-1, 3]$; range $[1/2, 7/2]$; minimum $(0, 1/2)$, maximum $(3/2, 7/2)$.

新域由2x∈[−2,6]。负的纵向因子交换极大与极小，分别映射(0,5)和(3,−1)。

| | |
|---|---|
| **M1** | 正确输入范围缩放 |
| **A1** | 域和值域 |
| **A1** | 两个正确极值点 |

### (c) · 3 marks

Domain $[-6, 6]$; range $[-1, 5]$. Both extreme outputs occur at non-negative inputs $0, 3$.

只用原f的[0,6]部分，然后反射到负横坐标；由于0和3已给最大、最小，且曲线连续，中间全部输出仍能达到。

| | |
|---|---|
| **M1** | 正确折叠定义域 |
| **A1** | 完整值域 |
| **B1** | 用两个极值位置解释 |

## Question 9

P3-C02-R09 · 9 marks

### (a) · 4 marks

$f^{-1}(x) = (x+1)/(x-1)$ on $x \neq 1$; $ff(x) = x$ for $x \neq 1$.

y(x−1)=x+1给x=(y+1)/(y−1)。原函数不能输出1，所以逆域排除1；它恰好是自身的逆。

| | |
|---|---|
| **M1** | 反解方程 |
| **A1** | 逆函数 |
| **B1** | 正确域 |
| **A1** | 合法复合恒等式 |

### (b) · 3 marks

The two graphs coincide; for example $(2, 3)$ lies on both and is not on $y = x$.

第一问已得f=f逆，整幅图重合。以f(2)=3给出明确不在对角线的公共点，即构成反例。

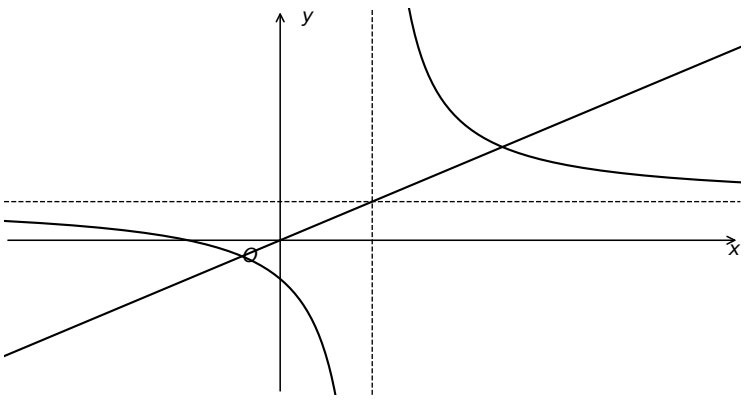
| | |
|---|---|
| **M1** | 由自身为逆说明两图重合 |
| **A1** | 有效离对角线公共点 |
| **B1** | 明确否定全称说法 |

### (c) · 2 marks

$(1+\sqrt{2}, 1+\sqrt{2})$ and $(1-\sqrt{2}, 1-\sqrt{2})$.

解(x+1)/(x−1)=x得$x^2-2x-1=0$；两根均不等于1，坐标纵横相等。

| | |
|---|---|
| **M1** | 联立并解二次式 |
| **A1** | 两精确坐标且域有效 |




Suggested sketch

## Question 10

P3-C02-R10 · 9 marks

### (a) · 3 marks

Range $[0, 4)$; $ff(1) = 2$.

第一段取[0,3]；第二段取[1,4)，合并[0,4)。f(1)=3采用第一段，再算f(3)=2。

| | |
|---|---|
| **M1** | 正确处理两段端点 |
| **A1** | 完整半开值域 |
| **A1** | 复合值 |

### (b) · 2 marks

No. Values approach 4 from below as $x \to 1^+$, but 4 is excluded and $f(1) = 3$.

靠近1右侧可输出比任何给定小于4的候选值更大的数，但4本身没有达到；不能把空心点当作函数值。

| | |
|---|---|
| **B1** | 未取得最大值 |
| **B1** | 空心端点及f(1)的解释 |

### (c) · 4 marks

$f^{-1}(x) = x - 2$ for $0 \le x \le 3$; $f^{-1}(x) = 5 - x$ for $7/2 \le x < 4$.

第二段现在输入(1,1.5]、输出[3.5,4)，与第一段输出[0,3]不重叠，因此限制后是一一映射。两支分别反解并保留输出域中的缺口。

| | |
|---|---|
| **M1** | 检查两输出段不重叠 |
| **M1** | 分别反解 |
| **A1** | 两支公式 |
| **A1** | 两个区间及缺口 |

## Question 11

P3-C02-R11 · 9 marks

### (a) · 3 marks

$fg(x) = ||x| - 6|$; $gf(x) = |x - 3| - 3$.

f(g(x))=||x|−3−3|；g(f(x))=||x−3||−3，由于内层非负，可去掉外绝对值。

| | |
|---|---|
| **M1** | 正确两次代入顺序 |
| **A1** | 第一复合 |
| **A1** | 第二复合的化简 |

### (b) · 4 marks

$x = -3$ or $x \geq 6$.

左边非负，故右边也须非负，先限x≤0或x≥6。x≥6时两式同为x−6，全部满足。x≤0再按−6分段：x≤−6无解，−6≤x≤0给x+6=−x，解−3。

| | |
|---|---|
| **M1** | 先限制右边非负的范围 |
| **M1** | 分别处理必要分段 |
| **A1** | 孤立解−3 |
| **A1** | 整段解x≥6 |

### (c) · 2 marks

No: $fg(0) = 6$ while $gf(0) = 0$.

相等方程有无限多个解也不能说明两函数相同；只需在同一合法输入0处给不等输出。

| | |
|---|---|
| **B1** | 正确反例值 |
| **B1** | 非恒等的结论 |

## Question 12

P3-C02-R12 · 9 marks

### (a) · 4 marks

Range $[0,3]$; $g^{-1}(x)=(3-x)^2-2$ for $0\le x\le 3$.

根号从0增加到3，故输出从3减到0；反解√(x+2)=3−y，输出范围确保右侧非负，再平方。

| | |
|---|---|
| **M1** | 从原域求完整范围 |
| **A1** | 值域 |
| **M1** | 合法反解并平方 |
| **A1** | 逆函数及其域 |

### (b) · 3 marks

$x=-1$.

先对外层用逆函数：g(x)=g逆(1)=2；再得x=g逆(2)=−1。核查g(−1)=2、g(2)=1。

| | |
|---|---|
| **M1** | 正确先撤销外层g |
| **M1** | 再次用逆函数 |
| **A1** | 有效解 |

### (c) · 2 marks

The range $[0,3]$ of the inner $g$ is contained in the outer domain $[-2,7]$.

只有内层输出能全部送入外层，复合域才保持原域；此处确有[0,3]包含于[−2,7]。

| | |
|---|---|
| **B1** | 正确两个范围 |
| **B1** | 包含关系与全域可复合的解释 |

## Question 13

P3-C02-R13 · 9 marks

### (a) · 4 marks

$fg(x) = \sqrt{9 - x^4}$; domain $[-\sqrt{3}, \sqrt{3}]$; range $[0, 3]$.

g输出$x^2$要在f域$[-3,3]$，因非负，等价$x^2 \leqslant 3$；所得四次根式仍能由0取到3。

| | |
|---|---|
| **M1** | 正确复合 |
| **M1** | 外层域对$x^2$的限制 |
| **A1** | 定义域 |
| **A1** | 值域 |

### (b) · 3 marks

$gf(x) = 9 - x^2$; domain $[-3, 3]$; range $[0, 9]$.

先做f需原域$[-3,3]$，其输出均可平方。平方抵消外根号，但不能据化简后的多项式把定义域扩展到实数。

| | |
|---|---|
| **M1** | 正确复合及化简 |
| **A1** | 保留原域 |
| **A1** | 完整值域 |

### (c) · 2 marks

$x = \pm\sqrt{2}$ and $x = \pm 2$, respectively.

第一式平方得$x^4=4$，实数根$\pm\sqrt{2}$均在第一复合域；第二式得$x^2=4$，两根在第二域。

| | |
|---|---|
| **M1** | 分别解两个复合方程 |
| **A1** | 两解集及域核对 |

## Question 14

P3-C02-R14 · 9 marks

### (a) · 3 marks

Range $[0,5]$; minima $(-2,0),(2,0)$.

绝对值最小0在原二次式零点$\pm2$；端点x=±3给5，中央峰在x=0仅高4，不是全局最大。

| | |
|---|---|
| **M1** | 两种候选极大位置比较 |
| **A1** | 值域 |
| **A1** | 两个最小点 |

### (b) · 3 marks

$-4<k<0$.

零点等价原图与水平y=−k相交；0<−k<4时四个交点，达到0或4时交点合并。

| | |
|---|---|
| **M1** | 用水平截线解释零点 |
| **M1** | 处理0与4两临界高度 |
| **A1** | 严格参数范围 |

### (c) · 3 marks

$k=-4$; $x=0,\pm2\sqrt{2}$.

高度4触中央峰，再与两侧外支相交；解$|x^2-4|=4$给$x^2=0$或8，域内共有三个不同根。

| | |
|---|---|
| **M1** | 临界高度的交点合并 |
| **M1** | 解绝对值方程并去重 |
| **A1** | k与三个精确根 |

## Question 15

P3-C02-R15 · 9 marks

### (a) · 3 marks

Line $y = x + 3$ with a hole at $(3, 6)$; range $\mathbb{R} \setminus \{6\}$.

约分得到x+3但原输入3仍不合法；在这条一一对应直线上删一个输入同时删对应输出6。

| | |
|---|---|
| **M1** | 正确约分但保留输入限制 |
| **A1** | 线与空心点 |
| **B1** | 值域缺6 |

### (b) · 4 marks

$qq(x) = x + 6$, domain $\mathbb{R} \setminus \{0, 3\}$, range $\mathbb{R} \setminus \{6, 9\}$.

内层先排除x=3，还须q(x)≠3，即x+3≠3，另排除0。复合化简为x+6，所以失去输出6与9。

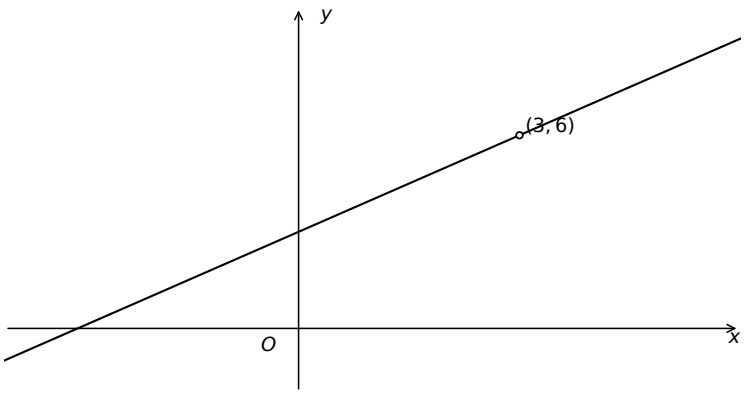
| | |
|---|---|
| **M1** | 同时检查两层定义域 |
| **A1** | 复合式 |
| **A1** | 完整域 |
| **A1** | 对应缺失值域 |

### (c) · 2 marks

$q^{-1}(x) = x - 3,\ x \neq 6$.

反解y=x+3为原输入y−3，但逆输入不能用原函数缺失输出6。

| | |
|---|---|
| **M1** | 反解直线 |
| **A1** | 正确逆函数及其域 |




Suggested sketch

## Question 16

P3-C02-R16 · 9 marks

### (a) · 3 marks

$k = -2$ or $k = 2$.

水平高度在$(-2,2)$内给三交点；达到极大或极小时一对根合并为转折点，另有对侧端点，故两点。

| | |
|---|---|
| **M1** | 比较内部水平截线与临界高度 |
| **A1** | 两个k值 |
| **B1** | 不重复计相切根 |

### (b) · 3 marks

Zeros $x = -\sqrt{3}, 0, \sqrt{3}$; height-2 points $(-2,2), (-1,2), (1,2), (2,2)$.

只把原图负值段关于横轴翻上去。零点由$x(x^2-3)=0$，端点和两转折点的绝对值都是2。

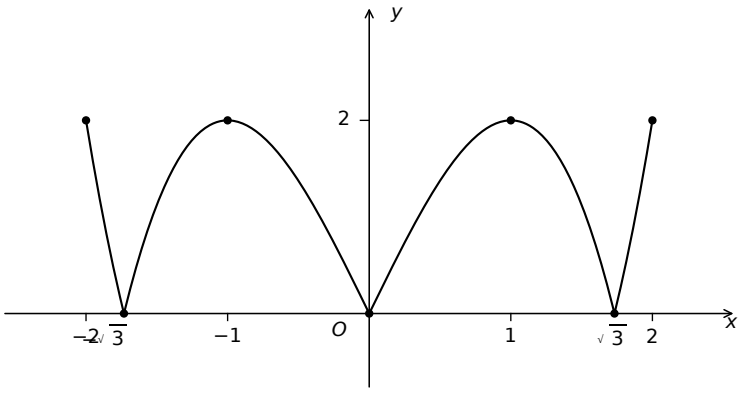
| | |
|---|---|
| **M1** | 正确反射负值部分 |
| **A1** | 三零点 |
| **A1** | 四个高度2的点与整体连线 |

### (c) · 3 marks

$0 < k < 2$.

等价f=k或f=−k，每条非临界截线各三根；k>0使两组不同。k=0仅三根、k=2仅四根，故端点排除。

| | |
|---|---|
| **M1** | 拆成两条水平截线 |
| **M1** | 数清互不重合的根 |
| **A1** | 严格参数区间 |




Suggested sketch

## Question 17

P3-C02-R17 · 11 marks

(a) · 4 marks

Range $[2,6]$; $f^{-1}(s)=18/s-3$ on $[2,6]$.

x+3在[3,9]内，倒数乘18得到输出[2,6]；反解18/(x+3)=s，得x=18/s−3。

| | |
|---|---|
| **M1** | 跟踪有限输入域到输出 |
| **A1** | 值域 |
| **M1** | 反解 |
| **A1** | 逆函数和域 |

(b) For the correction procedure $H(x)=f^{-1}(f(x)+1)$,

(i) · 3 marks

$3/5\le x\le 6$.

修正后读数必须满足2⩽f(x)+1⩽6。下限因f⩾2自动成立，上限要求18/(x+3)⩽5，故x⩾3/5，再与原域求交。

| | |
|---|---|
| **M1** | 检查修正读数落在逆函数定义域 |
| **M1** | 正确处理正分母不等式 |
| **A1** | 完整区间 |

(ii) · 4 marks

$H(x)=(15x-9)/(x+21)$; input $x=15/7$.

代逆函数得18/[18/(x+3)+1]−3，合并为(15x−9)/(x+21)。令其为1，得14x=30，且在有效域内。

| | |
|---|---|
| **M1** | 正确复合与通分 |
| **A1** | 最简式 |
| **M1** | 解输出方程并检查域 |
| **A1** | 有效输入 |

## Question 18

P3-C02-R18 · 10 marks

### (a) · 4 marks

Domain $[-9, 12]$; range $[-9, 3]$.

x/3必须在[−3,4]，所以横向放大3倍；纵输出负倍数−2交换范围两端，再减1，得[−9,3]。

| | |
|---|---|
| **M1** | 输入域变换 |
| **A1** | 定义域 |
| **M1** | 处理负倍数与平移 |
| **A1** | 值域 |

### (b) · 3 marks

Maximum $(0, 3)$; minimum $(6, -9)$.

原最小(0,−2)映成(0,3)，原最大(2,4)映成(6,−9)；负号交换两者性质。

| | |
|---|---|
| **M1** | 变换两个点 |
| **A1** | 两坐标 |
| **B1** | 正确极值性质 |

### (c) · 3 marks

$x = -6, 3, 9$.

h=−1等价f(x/3)=0，内层输入只能取原三个零点−2、1、3，故原输入各乘3。无需猜测图中其他高度的交点。

| | |
|---|---|
| **M1** | 目标高度转换到f的零水平 |
| **M1** | 正确反向横坐标映射 |
| **A1** | 三个解 |

## Question 19

P3-C02-R19 · 8 marks

### (a) · 3 marks

$F(x) = 9 - 3x$ for $x < 1$; $7 - x$ for $1 \le x \le 4$; $3x - 9$ for $x > 4$. Corners $(1, 6), (4, 3)$.

按1和4两地点分三段，逐个确定两个差值的符号。中段依然递减而不是水平，因为两运输权重不相同。

| | |
|---|---|
| **M1** | 在两个位置分段 |
| **A1** | 三个表达式和域 |
| **A1** | 两角点与正确三斜率 |

### (b) · 2 marks

$1 \le x \le 5$.

左段要求x≥1故没有左开段内部解；中段全满足；右段3x−9≤6给x≤5，与x>4合并。

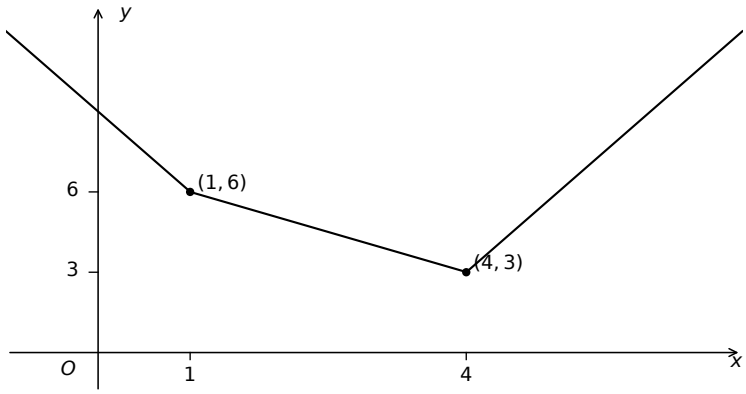
| | |
|---|---|
| **M1** | 各段解并结合对应范围 |
| **A1** | 两端与完整区间 |

### (c) · 3 marks

$F^{-1}(c) = (c + 9)/3$ for $c \ge 3$.

该限制只留严格递增右支，成本最小3且无上界；从c=3x−9反解即可。

| | |
|---|---|
| **M1** | 选择一一右支并确定成本域 |
| **A1** | 逆函数 |
| **B1** | 正确逆域 |




Suggested sketch

## Question 20

P3-C02-R20 · 8 marks

### (a) · 4 marks

$f^{-1}(x) = 1 + \sqrt{16 - x^2}$ on $[0, 4]$, with range $[1, 5]$.

原横坐标≥1，所以x−1非负；反解时取正平方根。原输出从4降到0，因此逆输入域[0,4]。

| | |
|---|---|
| **M1** | 合法平方并解原输入 |
| **A1** | 选择正确分支 |
| **A1** | 逆函数 |
| **B1** | 逆域与范围 |

### (b) · 4 marks

Only $((1 + \sqrt{31})/2, (1 + \sqrt{31})/2)$.

原图满足$(x-1)^2+y^2=16$，逆图满足$x^2+(y-1)^2=16$，相减得$y=x$。代回解$2x^2-2x-15=0$，负号根不在两个图的共同域；正号根合法。此理由来自本题两方程，不是普遍定理。

| | |
|---|---|
| **M1** | 写出两图方程并相减得y=x |
| **M1** | 解二次方程 |
| **A1** | 正确精确坐标 |
| **B1** | 排除非法根与完整性 |

## Question 21

P3-C02-R21 · 9 marks

### (a) · 4 marks

Intercepts $(0,3),(2,0)$; asymptote $y=4$ as $x\to-\infty$; range $[0,\infty)$.

x<2时函数4−2^x，左端趋4但小于4；x≥2时2^x−4无限增长。左支的极限4不妨碍右支实际达到4。

| | |
|---|---|
| **M1** | 正确反射原指数负值支 |
| **A1** | 两截点 |
| **B1** | 渐近线方向 |
| **B1** | 完整值域 |

### (b) · 3 marks

$0<k<4$.

拆成2^x=4+k与2^x=4−k。第一式k≥0时可解；第二式要求k<4；k=0两式同根，k=4第二式真数0无解。

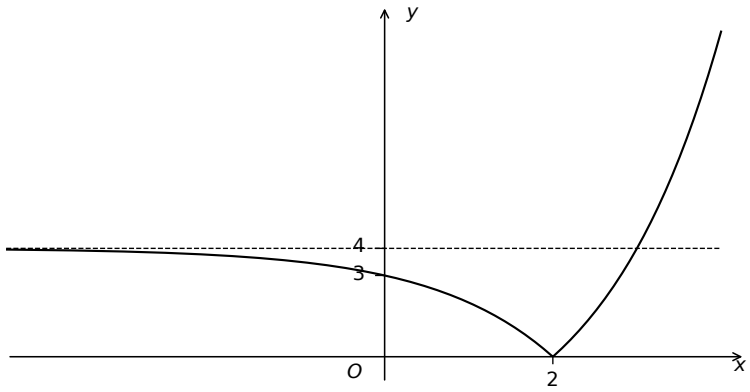
| | |
|---|---|
| **M1** | 两指数分支 |
| **M1** | 真数严格正的条件 |
| **A1** | 正确开区间 |

### (c) · 2 marks

$x=0$ or $x=\log_2 7$.

分别由2^x=1或7，得到0与$\log_2 7$；两真数均正，不需P3指数微分。

| | |
|---|---|
| **M1** | 正确两绝对值分支 |
| **A1** | 两精确解 |




Suggested sketch

## Question 22

P3-C02-R22 · 10 marks

### (a) · 4 marks

Asymptotes $x = 0, y = 1$; zero $(1, 0)$; range $[0, \infty)$.

先画1/x−1，再把负值部分翻上；在x<0与x>1均为1−1/x，但分属不同分支；0

| | |
|---|---|
| **M1** | 正确三段形状 |
| **A1** | 两渐近线 |
| **B1** | 零点 |
| **B1** | 完整值域 |

### (b) · 3 marks

$k > 0, k \neq 1$.

两候选输入为1/(1+k)与1/(1−k)。k=0时两候选重合；k=1时第二个不存在；其他正k均给两个合法非零输入。

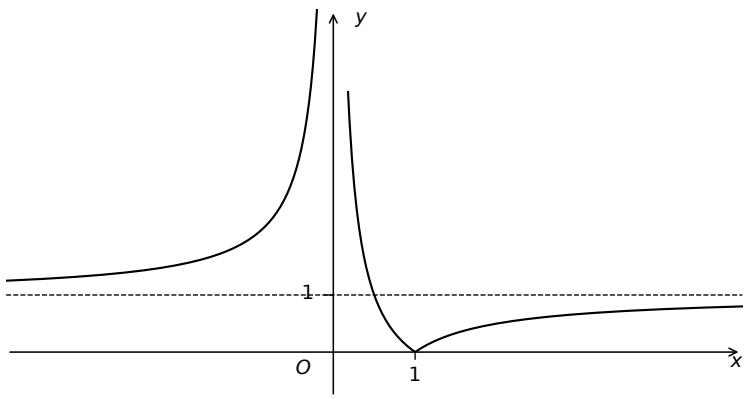
| | |
|---|---|
| **M1** | 解两个有理式分支 |
| **M1** | 检查重根和零分母例外 |
| **A1** | 完整参数集合 |

### (c) · 3 marks

$-1 < x < 0$ or $0 < x < 1/3$.

按负横坐标、正横坐标小于1和大于1分段；负区1−1/x>2给−12给01时函数<1无解。

| | |
|---|---|
| **M1** | 完整分段并按分母符号解不等式 |
| **A1** | 负区 |
| **A1** | 正区与排除0 |




Suggested sketch

## Question 23

P3-C02-R23 · 10 marks

(a) · 3 marks

$f^{-1}(x) = 2 + \sqrt{x-3}$ for $x \geq 3$.

f=(x−2)²+3，原横坐标在顶点右侧，输出≥3，取正分支求逆。

| | |
|---|---|
| **M1** | 配方并限制根号分支 |
| **A1** | 逆函数 |
| **B1** | 逆域 |

(b) · 3 marks

$g(2) = g(3) = 1$; range $[3/4, \infty)$.

g=x²−5x+7=(x−5/2)²+3/4。定义域包含顶点两侧的2和3，所以一一性丢失；顶点仍在域内，给最小3/4。

| | |
|---|---|
| **M1** | 配方或有效重复值 |
| **A1** | 非一一的具体证据 |
| **A1** | 完整值域 |

(c) · 4 marks

On $[2, 5/2]$: inverse $(5 - \sqrt{4x-3})/2$ for $3/4 \leq x \leq 1$. On $[5/2, \infty)$: inverse $(5 + \sqrt{4x-3})/2$ for $x \geq 3/4$.

以顶点5/2分割原输入范围。两支反解为5/2±√(x−3/4)；左支仅有原输入[2,2.5]，所以其输出域是[.75,1]，不能把两逆函数都写成同一全半轴。

| | |
|---|---|
| **M1** | 找两段最大单调输入区间 |
| **M1** | 反解两个根号分支 |
| **A1** | 左支公式及有限域 |
| **A1** | 右支公式及半轴域 |

## Question 24

P3-C02-R24 · 10 marks

(a) · 3 marks

$f(x) = 2 - 3|x|/2$ on $[-2, 2]$; range $[-1, 2]$.

顶点在(0,2)，左右斜率大小均3/2且向下，故表达式如此；图像连续覆盖−1到2。

| | |
|---|---|
| **M1** | 由图恢复斜率和绝对值结构 |
| **A1** | 正确函数 |
| **B1** | 完整值域 |

(b) · 3 marks

$f^{-1}(x) = (4 - 2x)/3$ on $[-1, 2]$.

右支是2−3x/2，一一递减；反解并使用这条支覆盖的输出区间[−1,2]。

| | |
|---|---|
| **M1** | 选择正确单调支 |
| **A1** | 逆式 |
| **B1** | 逆域 |

(c) · 4 marks

$x = \pm 4/9$.

外层f的零点是±4/3，但内层f只能输出[−1,2]，所以−4/3不可能。只解f(x)=4/3，得|x|=4/9。

| | |
|---|---|
| **M1** | 找外层的两候选零输入 |
| **B1** | 以内层值域排除−4/3 |
| **M1** | 解剩下的绝对值式 |
| **A1** | 两根 |

## Question 25

P3-C02-R25 · 11 marks

(a) · 3 marks

$h(x) = x/(x+4)$; range $[0, 1)$.

f可写1−2/(x+4)，所以f输出[1/2,1)，与g输入域恰匹配。代入g得2(x+2)/(x+4)−1=x/(x+4)，输出从0到趋近1。

| | |
|---|---|
| **M1** | 检查两步的范围匹配 |
| **A1** | 复合简式 |
| **A1** | 完整值域 |

(b) The complete calibration $h$ is used to recover the original non-negative input from an output reading $r$.

(i) · 3 marks

$h^{-1}(r) = \dfrac{4r}{1-r}$, $0 \le r < 1$.

h(x)=x/(x+4)，r(x+4)=x给x=4r/(1−r)；原输出范围作逆函数域。

| | |
|---|---|
| **M1** | 正确反解 |
| **A1** | 反函数 |
| **B1** | 定义域 |

(ii) · 2 marks

$4/3$.

代入(i)得(4×1/4)/(3/4)=4/3。

| | |
|---|---|
| **M1** | 代入完整逆校准 |
| **A1** | 输入 |

(c) · 3 marks

$h^{-1} = f^{-1}g^{-1}$, so $g$ must be undone first. Reading $1/4$ is not even in the domain $[1/2, 1)$ of $f^{-1}$.

输出属于最后一步g，正确先反演g。若对1/4直接用f逆，输入已不在f可实现输出范围内，算式即便能形式代入也不是合法逆校准。

| | |
|---|---|
| **M1** | 正确反序规则 |
| **B1** | 用1/4与f逆域给具体非法性 |
| **A1** | 完整错误解释 |

## Question 26

P3-C02-R26 · 9 marks

(a) · 4 marks

Range $(0,1)$; $f^{-1}(x)=\dfrac{x+2}{1-x}$ for $0<x<1$.

f=1−3/(x+1)，x>2时从0上升趋近1但不取端值；反解yx+y=x−2。

| | |
|---|---|
| **M1** | 范围端点与开闭 |
| **M1** | 反解 |
| **A1** | 逆表达式 |
| **B1** | 逆定义域 |

(b) For the composite $gf$, find

(i) · 3 marks

$gf(x)=\dfrac{x+1}{x-2}$, $x>2$.

f输入先需x>2，且其输出正，均可送入g；取倒数。

| | |
|---|---|
| **M1** | 正确复合顺序 |
| **A1** | 表达式 |
| **B1** | 定义域 |

(ii) · 2 marks

$(1,\infty)$.

gf=1+3/(x−2)，x>2时分式为任意正数，1不取到。

| | |
|---|---|
| **M1** | 正确改写与正值 |
| **A1** | 完整值域 |

## Question 27

P3-C02-R27 · 9 marks

(a) · 3 marks

$f^{-1}(x) = x + 1, x \geq 1.$

原域使f=x−1且输出⩾1，反解原输入。

| | |
|---|---|
| **M1** | 用限制去绝对值 |
| **M1** | 反解 |
| **A1** | 逆函数及域 |

(b) For $h$, find

(i) · 3 marks

Domain $x \leq 1$; $h(x) = 2 - x$.

内部输入3−x⩾2给x⩽1；在此域内|1−x|=1−x，所以加1为2−x。

| | |
|---|---|
| **M1** | 内部输入约束 |
| **M1** | 合法去绝对值 |
| **A1** | 域与式 |

(ii) · 3 marks

$h^{-1}(x) = 2 - x$, domain $x \geq 1$, range $x \leq 1$.

h在x⩽1上输出⩾1，反解同一线性式，交换域和值域。

| | |
|---|---|
| **M1** | 反解 |
| **A1** | 逆函数 |
| **B1** | 域与值域 |

## Question 28

P3-C02-R28 · 9 marks

(a) · 3 marks

Range $[1,3)\cup\{5\}$; one-to-one.

半开线段给[1,3)，孤立端点给5；各输出只对应一个输入，3、4不在值域。

| | |
|---|---|
| **M1** | 正确半开输出区间 |
| **A1** | 加孤立值 |
| **B1** | 一一性结论 |

(b) For the inverse function, find

(i) · 3 marks

$f^{-1}(x)=x-1$ for $1\le x<3$, and $f^{-1}(5)=2$.

每一段分别反解，保留孤立输入5，不填补3到5的空缺。

| | |
|---|---|
| **M1** | 连续分支反解 |
| **A1** | 孤立分支 |
| **B1** | 完整定义域 |

(ii) · 3 marks

$0\le x\le 1$.

需要x在原域且f(x)∈[0,2]；在半开线段f=x+1⩽2给x⩽1，x=2时输出5非法。

| | |
|---|---|
| **M1** | 复合输入条件 |
| **M1** | 排除特殊端点 |
| **A1** | 最终区间 |

## Question 29

P3-C02-R29 · 8 marks

(a) · 3 marks

$(0,7),(3,1),(5,5)$; $1 \le f(x) \le 7$.

分段为$7-2x$（$0 \le x \le 3$）和$2x-5$（$3 \le x \le 5$）。顶点最小值$1$，两端比较最大值$7$。

| | |
|---|---|
| **M1** | 正确分段图 |
| **A1** | 三点 |
| **B1** | 值域 |

(b) The domain is now restricted to $k \le x \le 5$, where $0 \le k < 5$, so that $f$ has an inverse.

(i) · 2 marks

$k = 3$.

若$k < 3$，两侧支段有重复输出，非一一映射；$k = 3$开始只保留严格递增支段。

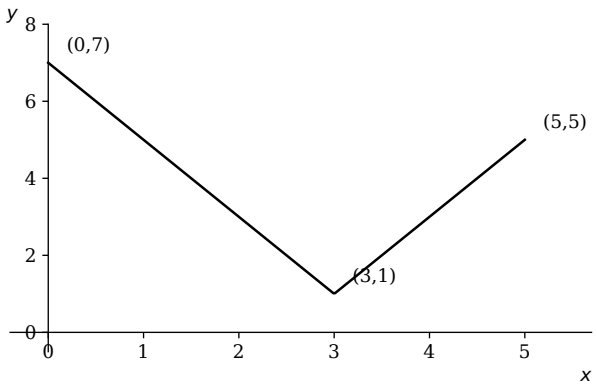
| | |
|---|---|
| **M1** | 用一一性判断 |
| **A1** | 最小k |

(ii) · 3 marks

$f^{-1}(x) = (x+5)/2$, $1 \le x \le 5$.

受限函数为$y = 2x - 5$，原定义域$[3,5]$、值域$[1,5]$；解出原输入并交换字母，反函数定义域取原值域。

| | |
|---|---|
| **M1** | 反解正确支段 |
| **A1** | 表达式 |
| **B1** | 定义域 |




Suggested sketch

## Question 30

P3-C02-R30 · 8 marks

(a) · 3 marks

$gf(x) = 1/(\sqrt{x+2}-1)$, $x \geq -2$, $x \neq -1$.

先要求根式$x+2 \geq 0$，再要求$f(x) \neq 1$，排除$x=-1$。两个限制均必要。

| | |
|---|---|
| **M1** | 正确复合顺序 |
| **A1** | 复合式 |
| **B1** | 全定义域 |

(b) For the composite function $fg$,

(i) · 3 marks

$x \leq 1/2$ or $x > 1$.

要求$1/(x-1)+2 = (2x-1)/(x-1) \geq 0$且$x \neq 1$。以$1/2, 1$分区间作符号判断。

| | |
|---|---|
| **M1** | 正确根式条件 |
| **M1** | 有理符号分析 |
| **A1** | 端点 |

(ii) · 2 marks

$x = 2$.

平方得$1/(x-1)+2 = 3$，所以$x=2$；属于前问定义域，代回根式为$\sqrt{3}$。

| | |
|---|---|
| **M1** | 求解复合方程 |
| **A1** | 合法输入 |